

Prediction of Vehicle Body and Tire Displacement Using a Suspension Recurrent Neural Network

Prem Yaragarla, py156

Department of Mechanical and Aerospace Engineering, Rutgers University

Abstract

Accurately modeling vehicle suspension is essential for ride comfort evaluation, structural design, and real-time control. Classical quarter-car models assume linear spring and damper behavior, but real suspension systems exhibit nonlinearities that these models cannot capture. This project presents a data-driven approach using a Gated Recurrent Unit (GRU) recurrent neural network to learn the input-output dynamics of a Quanser active suspension system directly from experimental data, bypassing the need for explicit system identification or knowledge of physical parameters. The model takes a sliding window of ISO 8608 road profile elevation as input and predicts both vehicle body and tire displacement. Training was performed across all five ISO 8608 roughness classes with 100,000 data points per class sampled at 1 kHz. Correlation was about 0.999 for all road classes. Body RMSE ranged from 0.0204 mm (Class A) to 0.0709 mm (Class E).

Problem Definition

Predicting vehicle suspension response to road excitation is a core problem in vehicle dynamics. Traditional approaches rely on physics-based models such as the quarter-car model, which require accurate knowledge of system parameters and assume linear behavior. This project addresses the problem of predicting both vehicle body and tire displacement simultaneously from road profile inputs using a data-driven approach. The experimental data was collected from a Quanser Active Suspension bench-scale system, which provides controlled, repeatable measurements. A GRU-based recurrent neural network was trained to learn the mapping from a time window of road elevation input to the corresponding displacement outputs, bypassing the need for explicit system identification and capturing nonlinear effects that linear models miss.

Materials and Methods

Dataset: The dataset originates from the Quanser Active Suspension benchmark. Road profiles were generated using the ISO 8608 standard PSD formulation with multisine signals across five roughness classes (A–E). Each input contains 100,000 data points sampled at 1 kHz, corresponding to a 2 km road traversed at 20 m/s. The measured displacements contain 400,000 data points from 4 periods; the first period was discarded as transient and the second period was used for training. Only unsuppressed (full signal) data was used, yielding 500,000 signal pairs for training. Data was normalized using global z-score scaling, which was more effective than MinMax scaling due to the factor-of-10 magnitude difference between Class A and Class E.

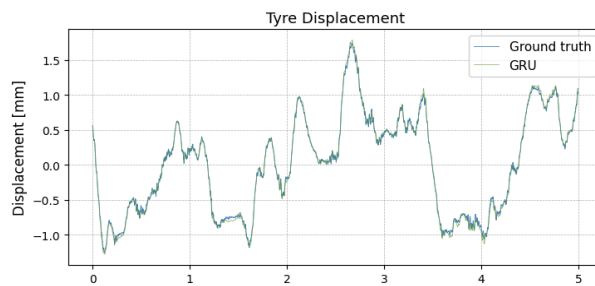
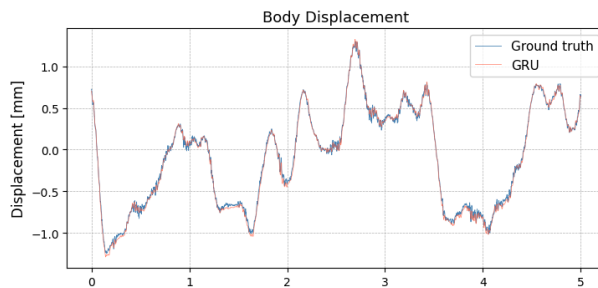
Neural network Architecture: The model is a GRU-based recurrent neural network designed to process sequential road profile data and predict the current body and tire displacement. The architecture consists of two stacked GRU layers followed by a three-layer fully connected (MLP) decoder, with ReLU activations between all layers. The final output layer produces two values

corresponding to body and tire displacement. This was done with the NAdam optimizer and MSE loss function, with ReduceLROnPlateau scheduler. The sequence length was 1,000 samples, stride was 5 samples, trained with 128 batch size for 300 epochs. The libraries used were PyTorch, SciPy, NumPy, Matplotlib and scikit-learn.

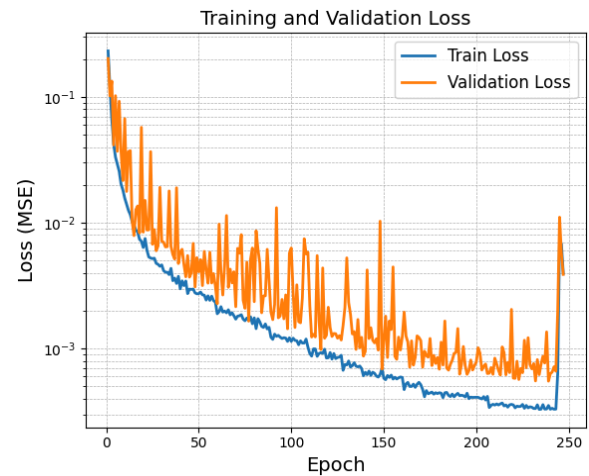
Results

Results after training show high accuracy from the model. The training and validation loss graph are shown below. When the loss spiked upwards, training was stopped. The error and correlation for each class is shown in the table below. The model has shown to predict rougher road profiles more accurately than the small deformations from class A. The predicted value of the body and tire displacement vs. road elevation is shown below for class C, which represents the majority of roads that will be driven on. The residuals are also shown.

Class	Body RMSE (mm)	Tire RMSE (mm)	Body Correlation	Tire Correlation
A	0.0204	0.0142	0.9970	0.9983
B	0.0234	0.0207	0.9993	0.9995
C	0.0293	0.0360	0.9996	0.9996
D	0.0482	0.0623	0.9998	0.9997
E	0.0709	0.1002	0.9999	0.9998



Plots showing body and tire displacement predictions against ground truth



Plot showing loss values during training

Key Takeaways

The GRU-based model successfully learns the input-output dynamics of the Quanser active suspension system directly from data, without requiring explicit system identification or knowledge of physical parameters. The model can produce physically reasonable predictions across all roughness levels, suggesting it captures nonlinear effects that linear models miss. The model is highly generalized due to training on all five road classes. One limitation is an inherent negative bias for smaller inputs, caused by Class E's larger magnitudes dominating the loss function during training.

Future Direction and Applications

The model can be improved by implementing physics-informed loss functions, such as energy balance constraints, to push the model towards physically sound predictions. However, testing is required since the real system's nonlinearities may conflict with simplified physics constraints. This work could also be extended to real-world road data, since the current training profiles are synthetic. In the future, this model can be implemented in active suspension systems where the ECU predicts suspension displacement and adjusts parameters for maximum performance or comfort.

LLM usage

Throughout this project, Claude was used to aid with coding, troubleshooting, and making changes to the model architecture. The most common use case was creating the skeleton of the code and adapting it to this specific problem. When training the model, I discussed the results with Claude to see what insights I could gain so that I could improve the architecture or understand why the model was acting a certain way.

Example prompts: “Find any parameters in the read me or other files such as spring and damper constants to create the physics informed model”

“What other suggestions do you think would help increase the accuracy of this model, should I introduce gradient tracking?”

“Will it work in real world situations when you won’t know what road class you’re on?”

Acknowledgments

I would like to thank Professor Vlassis for his guidance and instruction in this course. The experimental data used in this study, consisting of 100,000 data points per ISO 8608 roughness class sampled at 1 kHz, was obtained from the Quanser active suspension dataset hosted on Zenodo (<https://zenodo.org/records/17232645>).